Lecture 2 — Sep 29

Matrix times vector

By rows:

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 0 & 3 \\ 3 & 1 & 4 \end{bmatrix}\begin{bmatrix} 2 \\ 5 \\ 0 \end{bmatrix} =$$

$$= \begin{bmatrix} 1\cdot 2 + 1\cdot 5 + 1\cdot 0 \\ 2\cdot 2 + 0\cdot 5 + 3\cdot 0 \\ 3\cdot 2 + 1\cdot 5 + 4\cdot 0 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ 11 \end{bmatrix}$$

By columns:

$$2\cdot\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + 5\cdot\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + 0\cdot\begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 7 \\ 4 \\ 11 \end{bmatrix} \checkmark$$

In general:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \qquad x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

m rows, $\underline{\underline{n}}$ columns $\qquad$ $\underline{\underline{n}}$ entries

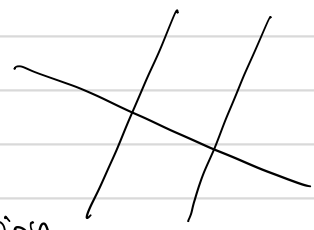
$$Ax = x_1\underbrace{\begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix}}_{\text{col } 1} + \cdots + x_n\underbrace{\begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix}}_{\text{col } n} = \begin{bmatrix} \circ \\ \circ \\ \vdots \\ \circ \end{bmatrix}$$

$\hookrightarrow$ ith entry: $a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{in}x_n$
$= \sum_{j=1}^{n} a_{ij}x_j$.

A singular system:

$$\left.\begin{aligned} 2u + v + w &= 5 \\ 4u + 2v + 2w &= 11 \end{aligned}\right\}$$
$$4u - 6v = -2$$

Inconsistent!
Parallel planes

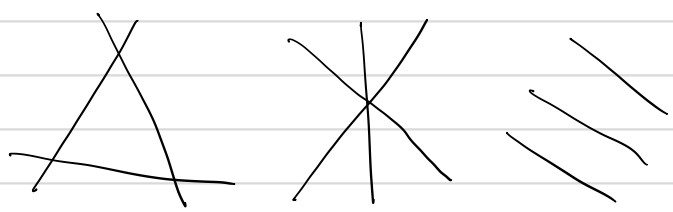


(a)

Elimination:

$$\begin{aligned} 2u + v + w &= 5 \\ R_2 \to R_2 - 2\cdot R_1: \quad \boxed{0 = 1} \\ 4u - 6v &= -2 \end{aligned}$$

Other possibilities:



(b) (c) (d)

Another singular system:

$$\begin{aligned} u + v + w &= 2 \\ 2u + 3w &= 5 \\ 3u + v + 4w &= 6 \end{aligned} \qquad (7)$$

No solution (b)

A line of soln's (e)

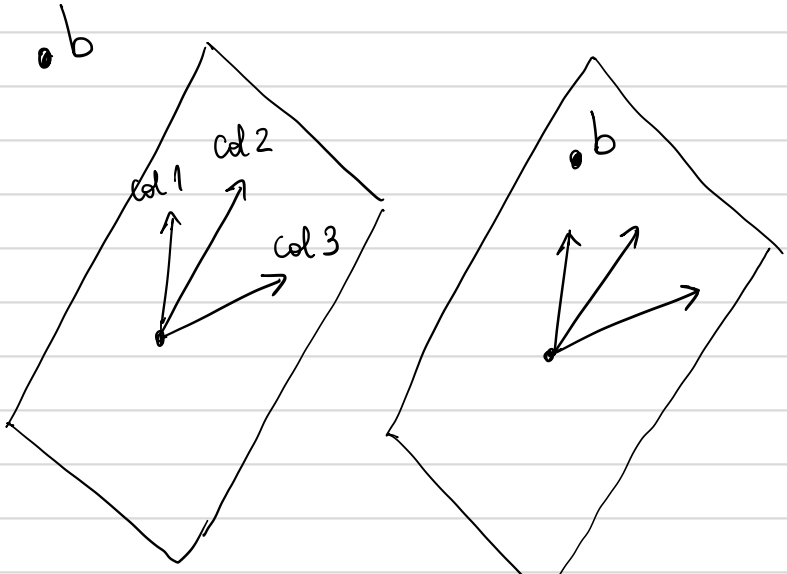
Column picture

$$u\begin{bmatrix}1\\2\\3\end{bmatrix}+v\begin{bmatrix}1\\0\\1\end{bmatrix}+w\begin{bmatrix}1\\3\\4\end{bmatrix}=b$$

→ 3 columns in a plane.

[col 2 = 3(col 1) − 2(col 1)]

Can solve only if b is in that plane.



no solution

∞ many solutions.

Parallelogram law:
Linear combinations of two vectors stay in the plane of the two vectors.

Elimination:

$$\begin{aligned} u+\ v+\ w&=2\\ 2u+\ \ \ \ \ 3w&=5\\ 3u+\ v+4w&=7 \end{aligned}$$

$\downarrow$ $R_2 \to R_2 - 2R_1$, $R_3 \to R_3 - 3R_1$

$$\begin{aligned} u+\ v+\ w&=2\\ -2v+\ w&=1\\ -2v+\ w&=1 \end{aligned}$$

$\downarrow$ $R_3 \to R_3 - R_2$

$$\begin{aligned} u+\ v+\ w&=2\\ -2v+\ w&=1\\ 0&=0 \end{aligned}$$

System in Echelon Form.
Leading variables: u, v
Free variables: w

To find all solutions, write the leading variables in terms of the free ones.

$$-2v = 1 - w \implies v = -\frac{1}{2} + \frac{w}{2}$$

$$u = 2 - \left(-\frac{1}{2} + \frac{w}{2}\right) - w = \frac{5}{2} - \frac{3}{2}w$$

For each $s \in \mathbb{R}$, get a soln w/ $w = s$:

$$\begin{bmatrix}u\\v\\w\end{bmatrix} = \begin{bmatrix}\frac{5}{2}-\frac{3}{2}s\\ -\frac{1}{2}+\frac{s}{2}\\ s\end{bmatrix}$$